



U1 Resource Guide

1 MATERIALS, CARD SIZES, MANIPULATIVES & PRINT INSTRUCTIONS

Merit Education Group™ — Intermediate Algebra Curriculum Unit 1: Variables and Expressions

2 HOW TO USE THIS GUIDE

This document lists every physical material needed for each lesson in Unit 1. Materials are organized by lesson and include: • **Printables** — card decks, game sheets, templates (with recommended paper size and color) • **Manipulatives** — physical items students handle • **Classroom supplies** — whiteboards, markers, timers, etc. • **Prep time** — estimated teacher preparation time **Quick-reference:** Lessons requiring **no special materials** (beyond worksheet + pen) are marked ✓ Minimal. Lessons needing card decks or cutouts are marked 📇 Cards needed.

3 MASTER PRINT TEMPLATE REFERENCE

Template	Size	Paper	Quantity	Used In
Game Card — Standard	3.5×2 in (9 per sheet)	Cardstock, white	1 set per 4-6 teams	L1–L5 games
Game Card — Large	5×3.5 in (4 per sheet)	Cardstock, white	1 set per class	Code Match
Score Tracker	Letter (8.5×11 in)	Copy paper	1 per class	L1–L5 games
Merit Time Scorekeeper	Letter (8.5×11 in)	Cardstock	1 per class	L1–L5 games
Lane Signs	Half-letter (5.5×8.5 in)	Cardstock, colored	2 per class	Speed Lane
Exit Ticket slip	3×5 in (4 per sheet)	Copy paper	1 per student	Every lesson

LESSON-BY-LESSON MATERIALS

Game: Translation Match | **Prep Time:** 20 min | Cards needed

Card deck contents: 8 pairs $\rightarrow$ word phrase + algebraic expression • a number plus 8 $\leftrightarrow n + 8$ • 5 less than a number $\leftrightarrow n - 5$ • twice a number $\leftrightarrow 2n$ • a number divided by 3 $\leftrightarrow n \div 3$ • the sum of a number and 6 $\leftrightarrow n + 6$ • the product of 4 and a number $\leftrightarrow 4n$ • a number decreased by 2 $\leftrightarrow n - 2$ • the quotient of a number and 5 $\leftrightarrow n \div 5$

Game: Expression Race | **Prep Time:** 10 min | Cards needed

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Card deck contents: 20 expressions flashed one at a time (e.g., $3x + 2$, $n - 7$, $2(x + 1)$). Students race to translate to verbal form.

L3 — Order of Operations (GEMDAS Basics)

Game: GEMDAS Challenge **Prep Time:** 5 min ✓ Minimal

Item	Specs	Qty	Notes
 Student Worksheet	Letter, copy paper	1 per student	
 Spanish Worksheet	Letter, copy paper	As needed	
 Scorekeeper	Letter, copy paper	1 per class	
 Timer	—	1	Game is on-screen only
 Whiteboards	—	1 per team	Teams show answers
No cut materials needed	—	—	All problems displayed on slides

L4 – GEMDAS Advanced Grouping & Exponents

Game: Code Match | **Prep Time:** 20 min | 👤 Cards needed

Item	Specs	Qty	Notes
 Student Worksheet	Letter, copy paper	1 per student	
 Spanish Worksheet	Letter, copy paper	As needed	
 Code Match cards	3.5×2 in, white cardstock	1 deck per team (16-24 cards)	See Game Files/ CodeMatch Master.md
 Scorekeeper	Letter, copy paper	1 per class	
 Timer	—	1	
 Whiteboards	—	1 per team	For tiebreakers

Card deck contents: 8-12 pairs of expression $\leftrightarrow$ value. Includes nested grouping & exponents.

L5 – Scientific Notation

Game: Power of Ten Relay | **Prep Time:** 10 min | 🧑🏫 Cards needed

Item	Specs	Qty	Notes
 Student Worksheet	Letter, copy paper	1 per student	
 Spanish Worksheet	Letter, copy paper	As needed	

Activity: Independent Practice | **Prep Time:** 5 min | ✓ Minimal

Item	Specs	Qty	Notes
 Student Worksheet	Letter, copy paper	1 per student	
 Spanish Worksheet	Letter, copy paper	As needed	
 Scorekeeper	Letter, copy paper	1 per class	
 Timer	—	1	
 Whiteboards	—	1 per team	Teams show formula answers
 Calculators	—	1 per student (optional)	Multi-step formula evaluation

L9 – Identifying Terms, Coefficients & Constants

Activity: Engagement Activity | **Prep Time:** 5 min | ✓ Minimal

Item	Specs	Qty	Notes
 Student Worksheet	Letter, copy paper	1 per student	
 Spanish Worksheet	Letter, copy paper	As needed	
 Scorekeeper	Letter, copy paper	1 per class	
 Timer	—	1	

L10 – Combining Like Terms

Activity: Small Group Practice **Prep Time:** 5 min ✓ Minimal

Item	Specs	Qty	Notes
 Student Worksheet	Letter, copy paper	1 per student	
 Spanish Worksheet	Letter, copy paper	As needed	
 Scorekeeper	Letter, copy paper	1 per class	
 Timer	—	1	

L11 — The Distributive Property

Activity: Independent Practice | **Prep Time:** 5 min | ✓ Minimal

Item	Specs	Qty	Notes
 Student Worksheet	Letter, copy paper	1 per student	

Item	Specs	Qty	Notes
 Timer	—	1	Essential — game is timed

Budget Balance Challenge format: Teams evaluate income and expense expressions to build a balanced budget (income – expenses $\geq$ 0). Fastest accurate budget wins.

Quiz 1 (Day 7) & Quiz 2 (Day 17)

Item	Specs	Qty
 Quiz printout	Letter, copy paper	1 per student
 Pencil	—	1 per student
 Timer	—	1 (15 min working time)

L18 — End-of-Unit Test (Day 20)

Item	Specs	Qty	Notes
 End-of-Unit Test printout	Letter, copy paper	1 per student	32 questions · 7 sections · 50 min
 Pencil	—	1 per student	
 Whiteboard	—	1	Display test directions
 Timer	—	1	50 min working time

5 CLASSROOM CONSUMABLES (WHOLE UNIT)

These are used across multiple lessons and should be stocked before starting Unit 1.

Item	Estimated Qty	Used In
Whiteboard markers (dry erase)	1 set per team	Every lesson
Index cards (3×5 in)	50	Exit tickets (L1-L18)
Scrap paper	1 ream	Practice problems
Pencils	1 per student	All lessons

6 CARDSTOCK PRINTING GUIDE

Card Type	Pieces Per Sheet	Sheets Needed	Lesson
Game cards (3.5×2 in)	9 per sheet	2-3 per deck	L1, L2, L4, L5, L6
Large game cards (5×3.5 in)	4 per sheet	1-2 per deck	L4
Lane signs (half-letter)	2 per sheet	1 sheet	L6

Total cardstock needed: Approximately 20-30 sheets of white + 5 sheets of colored cardstock for the full unit.

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QUICK-REFERENCE: LESSON MATERIAL NEEDS

Lesson	Game	Cardstock?	Manipulatives?	Special Items
L1	Translation Match	● 2-3 sheets	—	—
L2	Expression Race	● 1-2 sheets	—	—
L3	GEMDAS Challenge	✓ None	Whiteboards	—
L4	Code Match	● 2-3 sheets	—	—
L5	Power of Ten Relay	● 1-2 sheets	—	Index cards (powers of ten)
L6	Speed Lane	● 1-2 sheets	—	Lane signs
L7	Small Group Practice	✓ None	—	—
L8	Independent Practice	✓ None	Calculators (optional)	—
L9	Engagement Activity	✓ None	—	—
L10	Small Group Practice	✓ None	—	—
L11	Independent Practice	✓ None	—	—
L12	Engagement Activity	✓ None	—	—
L13	Large Group Practice	✓ None	—	—
L14	Independent Practice	✓ None	—	—
L15	Small Group Practice	✓ None	—	—
L16	Engagement Activity	✓ None	Whiteboards	—
L17	Budget Balance Challenge	✓ None	Calculators	—
Quiz 1	—	✓ None	—	—
Quiz 2	—	✓ None	—	—
L18	End-of-Unit Test	✓ None	Whiteboards	—

Legend: ✓ None = no printing/cutting needed | ● = cardstock recommended

